

Robust and Optimal Contention Resolution without Collision Detection

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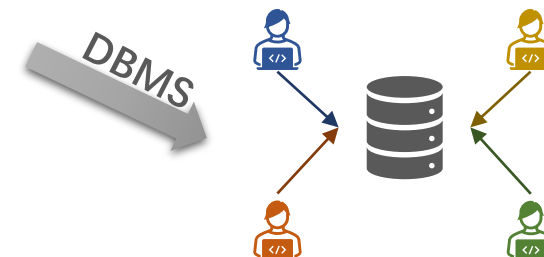
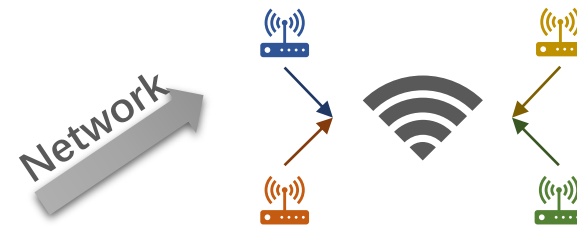
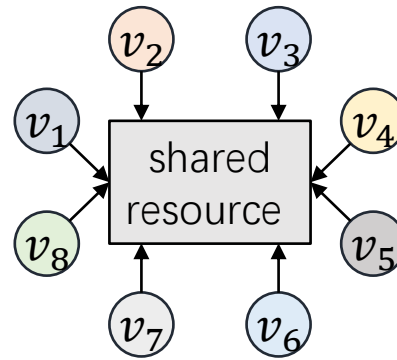
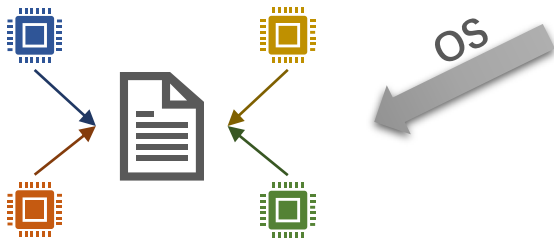
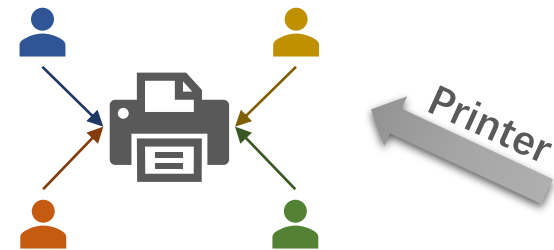
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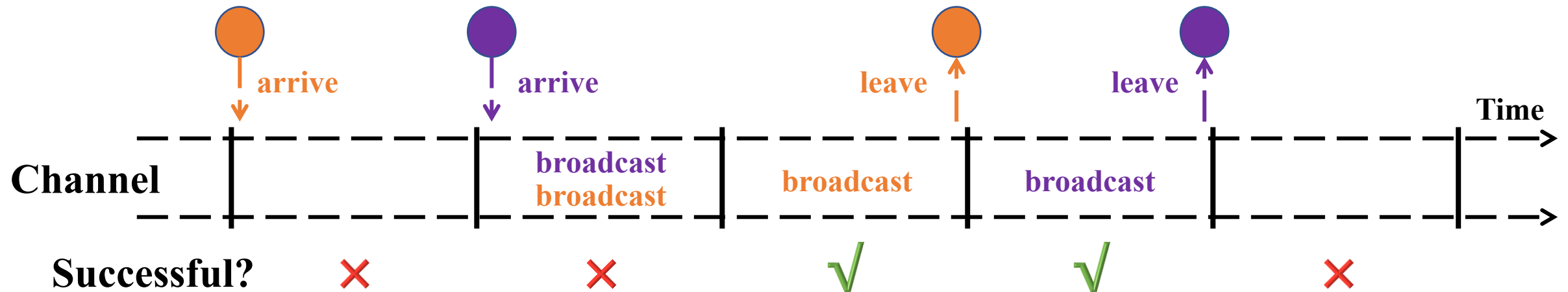
Contention resolution problem

- Many nodes want to access a shared resource.
- Successful access must be **mutual exclusive**.



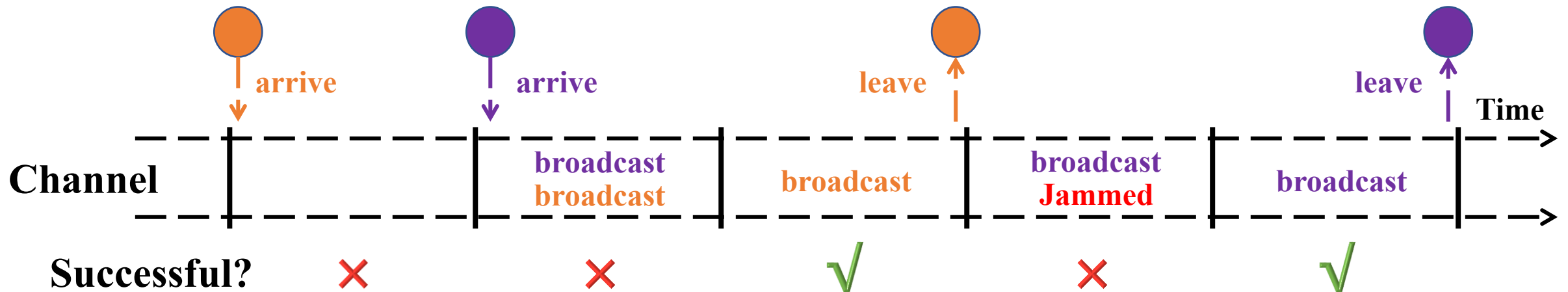
Formal definition

- Time is divided into **synchronized** slots.
- n nodes each arrives at the **beginning** of some slot. (Controlled by an adversary)
- In Each slot, each node either **broadcasts** or **idles**.
- Successful broadcast happens if **exactly one** node broadcasts in a slot.
- A node leaves **immediately** after it successful broadcasts its message.



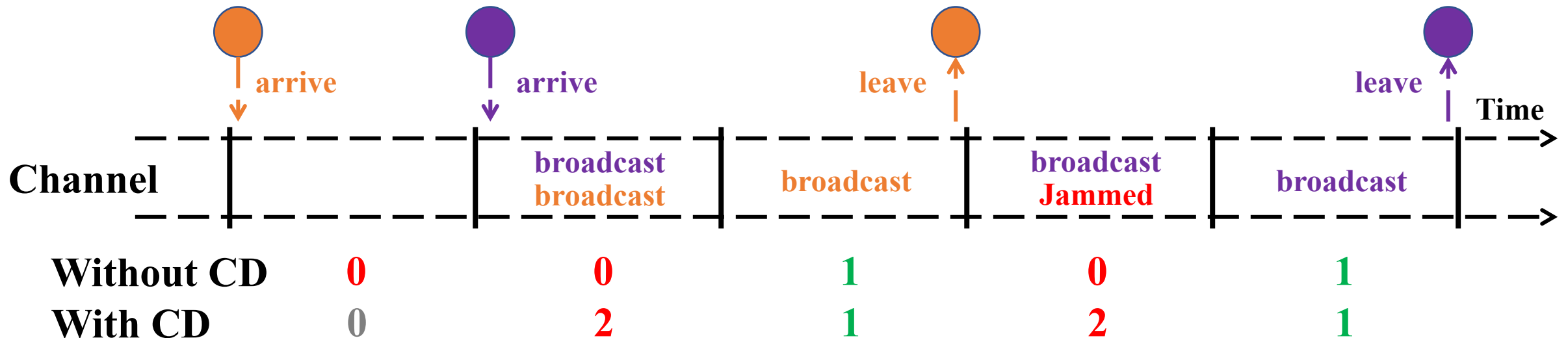
Robustness

- A jammed slot is always unsuccessful. (An adaptive adversary controls)
- The adversary can jam at most d slots.



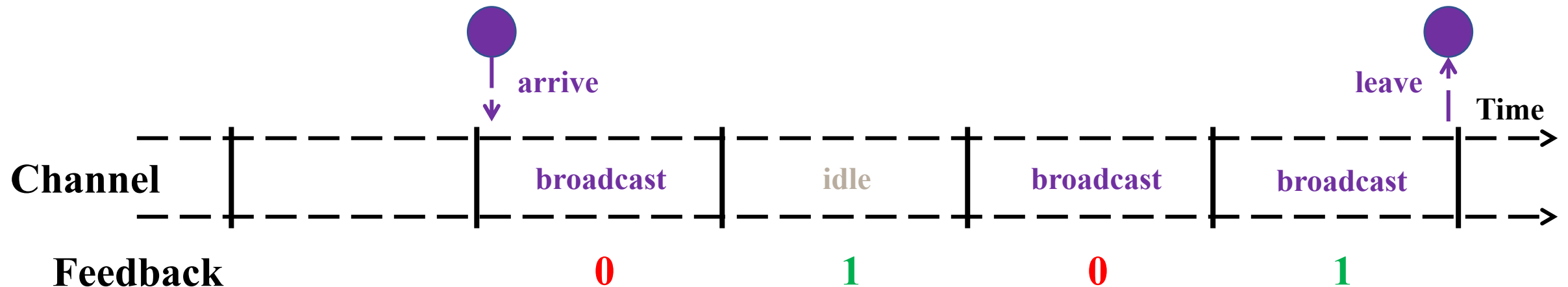
Feedback

- Each node gets **feedback** from the channel at each slot.
- Without Collision Detection, the feedback is $\begin{cases} 0 \text{ (failed)} \\ 1 \text{ (success)} \end{cases}$
- With Collision Detection (CD), the feedback is $\begin{cases} 0 \text{ (empty)} \\ 1 \text{ (success)} \\ 2 \text{ (collision)} \end{cases}$



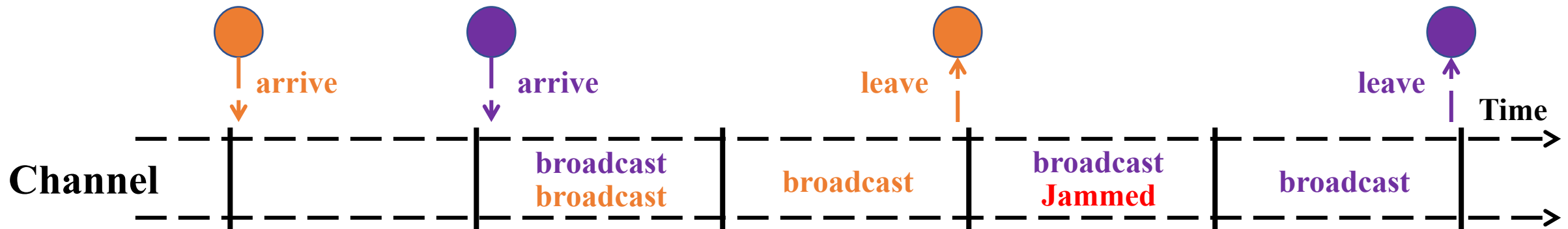
The view of one node

- No global clock.
- Behavior of each node at each slot: a **function** of feedback and its own behavior from previous slots.
- **Goal:** design good **function** (or simply called algorithm).



Measurements of algorithm

- **Variables:** Number of nodes n . Number of jammed slots d . $n = 2 \quad d = 1$
- **Time complexity:** Number of occupied slots. $T(n, d) = 5$
- **Energy complexity:** Average broadcast attempts. $E(n, d) = \frac{5}{2}$
- *Randomized algorithm: with probability at least $1 - \frac{1}{n}$



Our results

- $T(n, d) = O(n \log n + d)$ **Optimal** * [Chen, Jiang, Zheng, PODC 21]
* gives an algorithm with **optimal** $T(n, d)$, but with **polynomial** $E(n, d)$.
- $E(n, d) = O(\log^2 n + \log^2 d)$ **Optimal** *

* Lower bounds are proved under $T(n, d) = O(f(n) + d)$

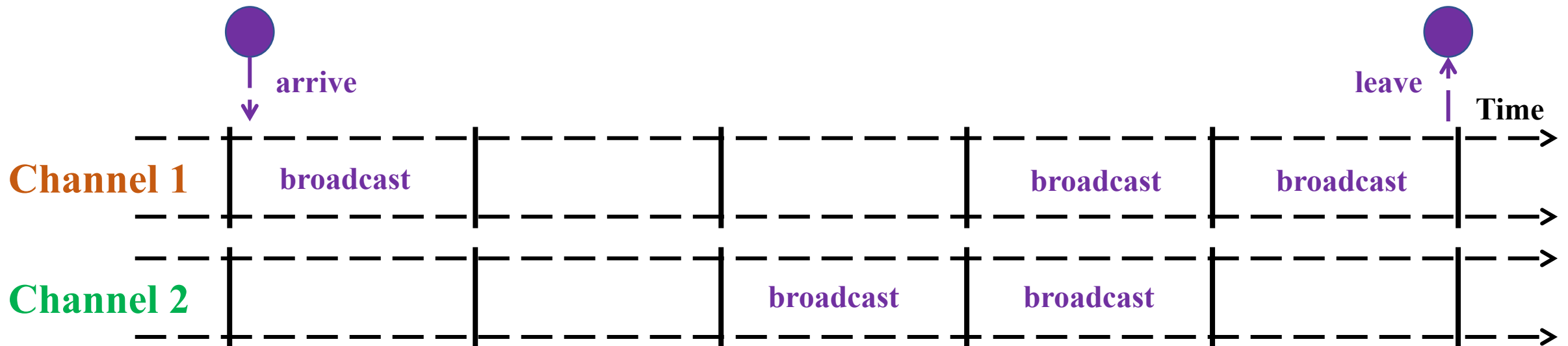
Static case: all nodes arrive in the same slot

- $T(n, d) = O(n + d)$ **Optimal**
- $E(n, d) = O(\log^2 n + \log^2 d)$ **Optimal on d**

Algorithm idea 1: two channels

- Summarize the idea in [BK KP 20, CJZ 21] and get a general framework to implement two channels by one channel.

* Not trivial since there are no global clock, nodes cannot distinguish odd and even slots.



Algorithm idea 2: contention

Definition (contention of a slot) : Sum of broadcast probability of all nodes in this slot. (jammed slots has contention ∞)

Lemma 1: When contention is small, the slot is successful with probability $\Theta(\text{contention})$.

- Contention between $\left[\Omega\left(\frac{1}{\log n}\right), O(1) \right]$ is good. $\Omega\left(\frac{1}{\log n}\right)$ success probability $\rightarrow$
 $O(n \log n)$ slots are enough to generate n successes.

Heavy slots has contention $\Omega(1)$

Light slots has contention $O(1)$ (and at least one node in the system)

Bound $\#\{\text{heavy slots}\} + \#\{\text{light slots}\}$

Algorithm idea 3: exponential backoff

Algorithm “*h*-backoff”: send with probability $\frac{h(i)}{i}$ at the i -th slot.

Lemma 2: If all nodes run $O(1)$ -backoff, there are at most $O(d + n \log n)$ heavy slots.

Idea: Each nodes contributes $O(\log t)$ contention to the first t slots.

Lemma 3: If all nodes run $O(\log n)$ -backoff, then $\#\{light\ slots\} = O(\#\{heavy\ slots\})$.

Idea: Each nodes u succeed in light slot with probability

$$\Pr[u \text{ broadcasts}] \cdot \Pr[\text{other nodes idel}] = \frac{\log \#\{heavy\ slots\}}{\#\{heavy\ slots\}} \cdot constant$$

In $\#\{heavy\ slots\}$ it succeeds w.h.p. Take union bound on all nodes.

Algorithm

$O(1)$ -backoff in **Channel 1**; $O(\log n)$ -backoff in **Channel 2**.

Time complexity: count the number of the following three types of slots:

- **Heavy slots in the first channel:** bounded by **Lemma 2**.
- **Light slots in the first channel with $\Omega\left(\frac{1}{\log n}\right)$ contention:** bounded by **Lemma 1**.
- **Other slots:** are light slots in the second channel, bounded by **Lemma 3**.

Energy complexity: $O(\log^2 T(n, d)) = O(\log^2 n + \log^2 d)$

Algorithm in static case

- **Fact 1:** the following algorithm from [BKPP 20] has $T(n, d) = O(n + d)$ in static case (but has large energy cost).

Algorithm 1:

- $O(1)$ -backoff in **Channel 1**; $O(\log n)$ -backoff in **Channel 2**.
- When seeing success in **Channel 2**, double broadcast probability.
- **Fact 2:** our dynamic algorithm (**Algorithm 2**) has $T(n, d) = O(n \log n + d)$ and good energy cost.

* Use **Algorithm 1** until $d > \tilde{n} \log \tilde{n}$, then use **Algorithm 2**.

* $\tilde{n}$ current number of nodes in the system

Algorithm in static case

Use **Algorithm 1** until $d > \tilde{n} \log \tilde{n}$, then use **Algorithm 2**.

- How can a node know when $d > \tilde{n} \log \tilde{n}$?
- **Claim 1 (approximate $\tilde{n}$):** sending probability in **Channel 1** $\approx 1/\tilde{n}$.
 - *Channel 2 controls the sending probability.*
- **Claim 2 (approximate d):** use current time slot t .
 - *Running time of Algorithm 1 is $O(n + d)$.*
 - *n can be removed after the first success in Channel 2.*

Contention resolution: related works

Harder



	With Collision Detection	Without Collision Detection
Without jamming	-	$T(n) = O(n)$ $E(n) = O(\log^2 n)$ [Bender, Kopelowitz, Kuszmaul, Pettie, 20]
With jamming (linearity on d)	$T(n, d) = O(n + d)$ $E(n, d) = O(1)$ [Chang, Jin, Pettie, 18]	$T(n, d) = O(n \log n + d)$ $E(n, d) = O(\log^2 n + \log^2 d)$ Our result

Open questions:

- Besides sending, also consider **listening** as an energy cost. [Chang, Kopelowitz, Pettie, Wang, Zhang, 18]
- Find a trade-off between n, d and still keep optimal energy cost.

Thank you!